

Pahamify: Making college entrance easy with exam prep features and AI insights on Google Cloud

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Pahamify migrated to automatically handle traffic spikes when hundreds of thousands of students participate in mock exams and data on BigQuery AI-driven learning

About Pahamify

Founded in 2018, Pahamify is an edtech platform that combines quality content and gamification to make learning fun. The mobile and web app delivers live classes, learning videos, practice quizzes, and mock examinations to help Grade 12 high school students prepare for the national entrance examination. More than 2.5 million students have downloaded the app from Apple and Google Play store.

Industries: Education

Google Cloud results

- Analyzes production data in 10 seconds with BigQuery to support real-time decision making
- Blends multiple data sources to find valuable insights on Looker Studio
- Operates AI data pipelines at scale to help students improve scores on mock examinations

Autoscale GKE to 10X its users

Location: Indonesia

Google Marketing Platform

Google Cloud (<https://cloud.google.com/>) (about),

Looker Studio

BigQuery (<https://cloud.google.com/bigquery/>) (data-studio/)

Vision AI (<https://cloud.google.com/vision/>)

(<https://marketingplatform.google.com/about/r>

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Google Marketing Platform

(<https://marketingplatform.google.com/about>)

Looker Studio

(<https://marketingplatform.google.com/about/data-studio/>)

Google Analytics

(<https://marketingplatform.google.com/about/resources/analytics-product-overview/>)

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*—Edria Albert, Chief
Technology Officer and
Co-Founder, Pahamify*

Admissions to Indonesian universities are fiercely competitive, with only one in four students securing

themselves a place. It is common for students to apply to universities and choose their majors based on the scores of their final year computer-based written exams (UTBK). The Pahamify (<https://pahamify.com/>) ed-tech platform offers study materials such as learning videos and quizzes, helping seven out of 10 students enter their school of choice.

When schools were closed during the COVID-19 pandemic, students across Indonesia worried about how they could study independently for the entrance exams. To help prepare them, Pahamify offered free live tutoring and mock exams. When the company discovered a 10X increase in users within two months, it migrated to Google Cloud (<https://cloud.google.com/>) to keep its digital resources uninterrupted, despite the sudden spikes in traffic.

According to Edria Albert, Chief Technology Officer and Co-Founder at Pahamify, the pandemic was both a game changer and a challenge for digital learning. Covid-19 changed the education landscape. Despite this, not all schools had the IT resources to set up online courses quickly.

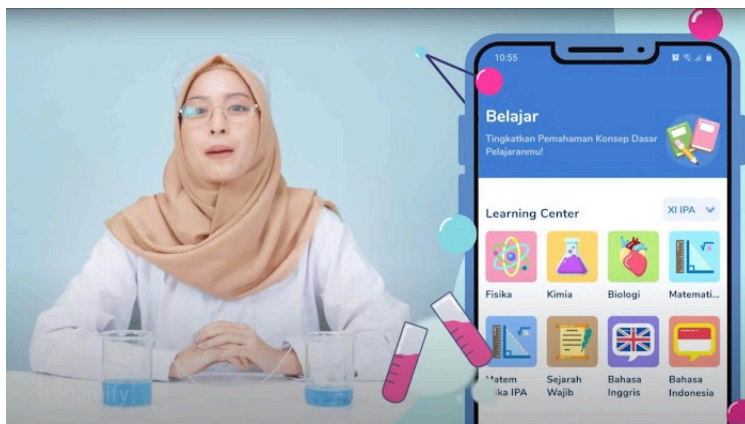
"Our platform powered by Google Cloud enables more than 2 million students access to quality exam prep content and teacher interaction in live classes," says Albert. "In addition, we can develop new features quickly to meet the needs of students, such as the Pegasus adaptive learning platform and the Ask Mipi AI question solver."

Pahamify uses Google Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine>) (GKE) with its automated features, such as provisioning and autoscaling, as well as BigQuery

(<https://cloud.google.com/bigquery>) and Looker Studio

(<https://lookerstudio.google.com/c/>) to improve decision making and learning performance for business teams and students.



Delivering a smooth learning experience with scheduled autoscaling

The demand for Pahamify varies with the academic year, as with any online learning platform. Typically, traffic spikes during the examination season, from thousands of users, to tens of thousands almost overnight. This is because it is common for students to take mock exams during this time to test their preparation. In 2021, 200,000 unique users took mock examinations on Pahamify, constituting a quarter of the students who took the UTBK, Indonesia's annual college entrance exam held by and for state universities.

Because the traffic pattern on GKE is fairly predictable, Pahamify increases its availability by leveraging schedule-based autoscaling. By prescaling ahead of the anticipated load, cloud resources are readily available for traffic spikes without the engineers wasting time on initialization.

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Educational and
Cultural Data
and Statistics.
With Looker
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Data insights at scale to improve learning experience and decision making

Pahamify's business runs on data. Subscribers should be able to access all their subscription features when logged in. Such account data is stored in the backend system. Each time a student completes a mock examination, they can check their ranking on Pahamify's national ranking leaderboard. They can then use this leaderboard to get an idea of where they stand among their peers.

Internally, Pahamify can use this data to make better real-time decisions. Product managers, for example, use data to learn about their users and improve their products such as self-learning, test simulation, live streams, and platform products.

In the past, Pahamify connected its business intelligence tool directly to its production database. When someone clicks on the dashboard, it sends a query to the underlying data. As a result of the growing number of users, queries overloaded the production database, which resulted in slow performance or even loss of service.

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"In BigQuery, we can analyze our production data within 10 seconds and integrate it with other data sources, such as Google Analytics (<https://analytics.google.com/analytics/web/provision/#/provision>)

and offline data sources such as Sheets or CSV files," says Albert. "With Looker Studio, we can display real-time data insights in an easy to use dashboard without requiring much assistance from engineering."

Product managers can see key performance indicators on how users interact with their products by looking at the dashboard. There is a greater interest in mock examinations when exam season is close, and with the data available, Pahamify is able

to curate and offer packages for different budgets to increase pickup rates. The data available also supports the business' growth. For example, as students' in-app time decreases during breaks and holidays, product managers can create more 'edutainment' videos to attract them.

Combining AI and human intelligence to close learning gaps



Online learning makes it difficult for students to receive feedback from their teachers. In the UNICEF U-Report Indonesia

(<https://indonesia.ureport.in/opinion/4283/>), 38% of students cited the lack of guidance from teachers as one of the challenges when studying at home. Based on the fact that students want feedback, Pahamify built an AI-based adaptive learning feature called Pegasus in three months and launched it in January 2022.

Using Pegasus, Pahamify analyzes terabytes of data collected during live classes and mock

examinations using BigQuery. It uses machine learning algorithms designed by its engineers and teachers to help students plan their studies effectively with recommended content, allowing them to improve their performance.

Pahamify also uses AI to help students solve problems. If students encounter difficult questions, they can take a photo of the question and submit it using the Ask Mipi question solver on the Pahamify app. Vision AI (<https://cloud.google.com/vision>) recognizes text in images through Optical Character Recognition and searches through the Pahamify question bank for a solution. The question solver then shows students relevant learning videos on how to solve the problem.

"Personalized learning is the key to the future of education. Quality content is beneficial to students' learning, but many rely on teachers for feedback and motivation," says Albert. "With Google Cloud, we can innovate on the learning experience with data insights and AI, so students have the resources and content they need to succeed."